

Supporting Information

Reprocessable Polypropylene Covalent Adaptable Networks via Free-Radical Reactive Processing with Vinyl Aromatic Additives

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Introduce resonance stabilization during radical-based reactive processing of PP

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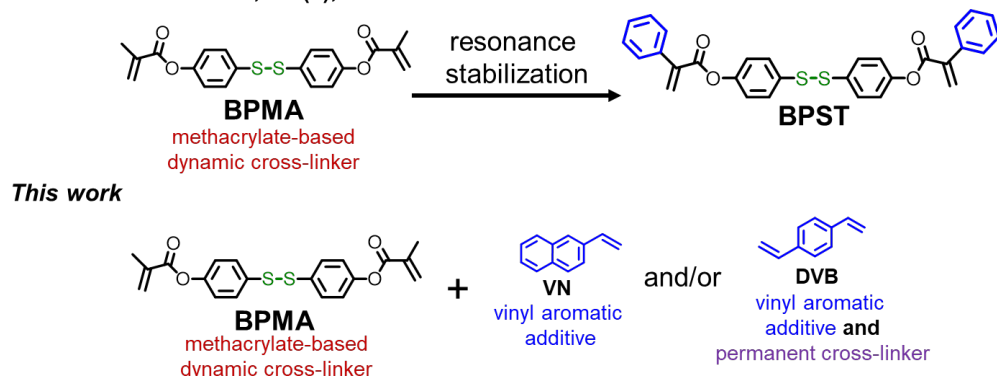


Figure S1. Approaches for introducing resonance stabilization during radical-based reactive processing of PP. Our previous study achieved resonance stabilization by modifying the structure of BPMA, a methacrylate-based cross-linker, to include phenyl groups (resulting in BPST, a phenylacrylate-based cross-linker).^{S1} In this study, we instead aimed to use vinyl-containing aromatic compounds, like 4-vinylnaphthalene (VN) and divinyl benzene (DVB), as stabilizing additives during radical-based cross-linking of PP with a methacrylate-based dynamic cross-linker.

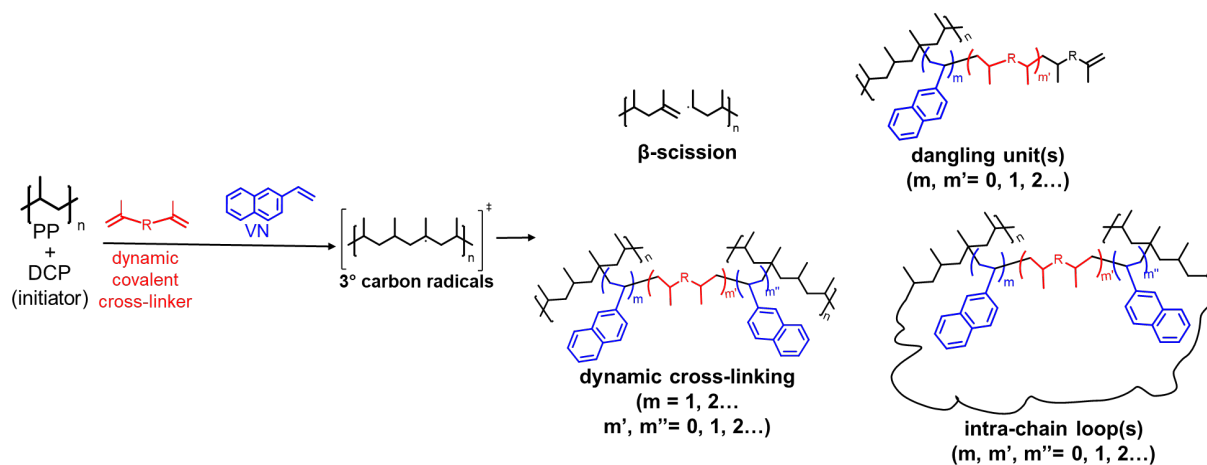


Figure S2. Possible reactions during radical-based reactive processing of PP homopolymer with dynamic covalent cross-linker, initiator, and VN at 180 °C.

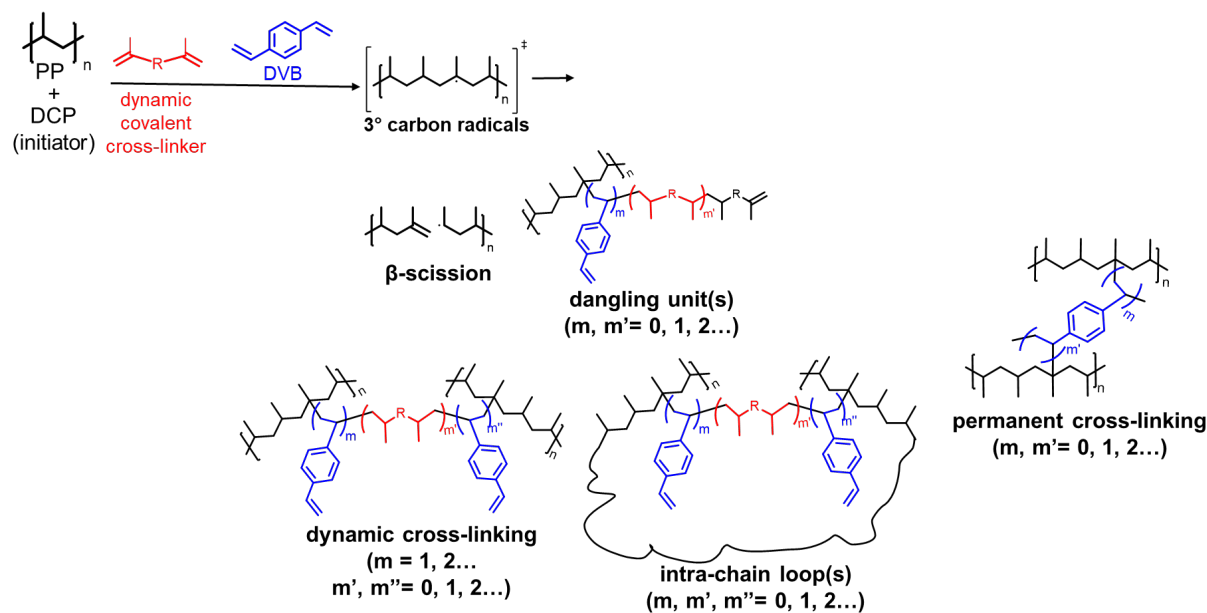


Figure S3. Possible reactions during radical-based reactive processing of PP homopolymer with dynamic covalent cross-linker, initiator, and DVB at 180 °C.

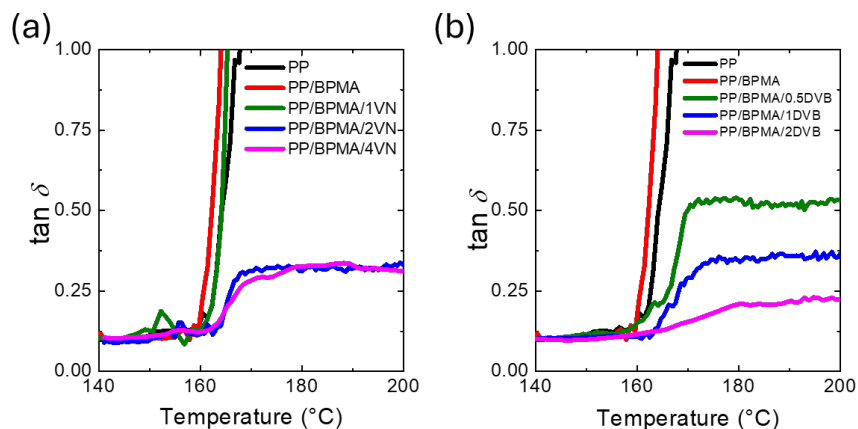


Figure S4. $\tan \delta$ (E''/E') as a function of temperature for neat PP and 1st molds of reactively processed PP samples. Reactively processed PP samples were made at 180 °C using 1.0 wt% DCP and 4.0 wt% BPMA along with (a) 0, 1, 2, or 4 molar equivalents of VN or (b) 0, 0.5, 1, or 2 molar equivalents of DVB with respect to BPMA. Data for the PP/BPMA sample is taken from ref. S1 (referred to as “PP12/BPMA4-1” in that work).

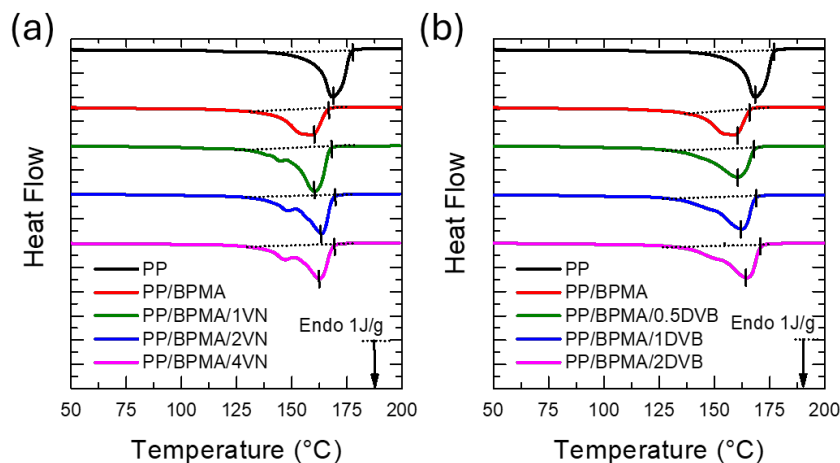


Figure S5. DSC thermograms for neat PP and 1st molds of reactively processed PP samples with $T_{m,peak}$ and $T_{m,endpoint}$ labeled. Reactively processed PP samples were made at 180 °C using 1.0 wt% DCP and 4.0 wt% BPMA along with (a) 0, 1, 2, or 4 molar equivalents of VN or (b) 0, 0.5, 1, or 2 molar equivalents of DVB with respect to BPMA. Data for the PP/BPMA sample is taken from ref. S1 (referred to as “PP12/BPMA4-1” in that work).

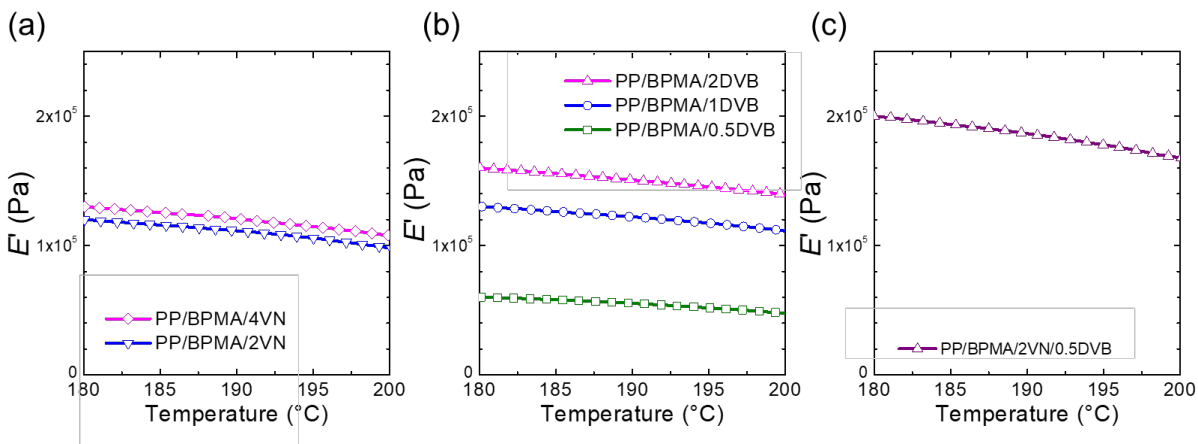


Figure S6. Non-normalized E' in the rubbery plateau as a function of temperature for 1st molds of reactively processed PP samples. Reactively processed PP samples were made at 180 °C using 1.0 wt% DCP and 4.0 wt% BPMA along with (a) 2 or 4 molar equivalents of VN, (b) 0.5, 1, or 2 molar equivalents of DVB, or (c) 2 molar equivalents of VN and 0.5 molar equivalent of DVB with respect to BPMA.

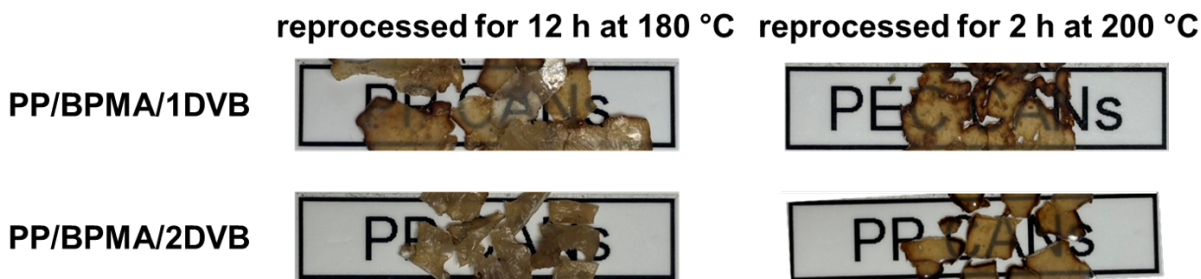


Figure S7. Unsuccessful attempts to produce consolidated 2nd mold samples from cut pieces of 1st mold samples of PP/BPMA/1DVB and PP/BPMA/2DVB with compression molding for 12 h at 180 °C or 2 h at 200 °C.

Table S1. KWW function parameters fitted from stress relaxation curves at 180 °C for 1st mold PP CANs

	Stress relaxation analysis fitted parameters ^{S2}			
	τ^* (s)	β	$\langle\tau\rangle$ (s)	R^2
PP/BPMA/2VN	59	0.55	100	0.98
PP/BPMA/0.5DVB	18	0.78	20	0.98
PP/BPMA/2VN/0.5DVB	165	0.59	254	0.99

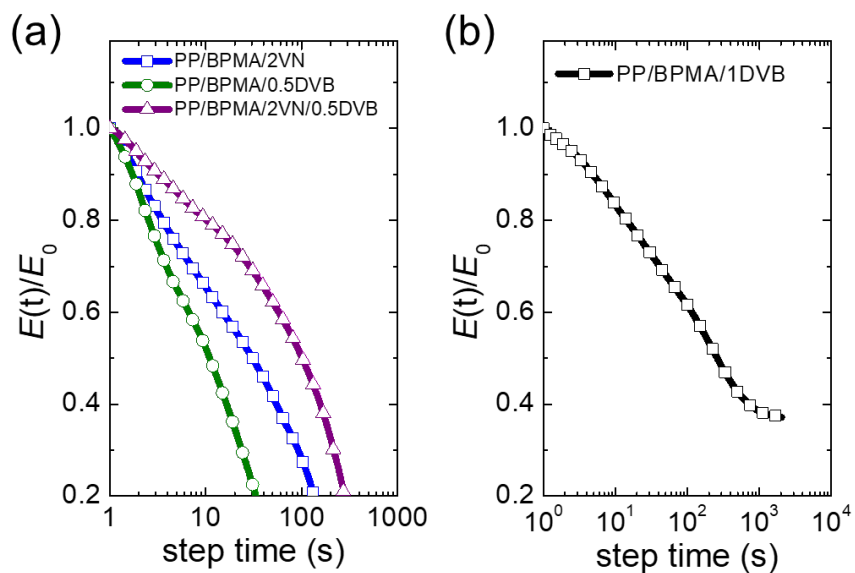


Figure S8. Normalized stress relaxation curves taken at 180 °C of (a) 1st mold PP CANs and (b) 1st mold PP/BPMA/1DVB, a permanent PP network.

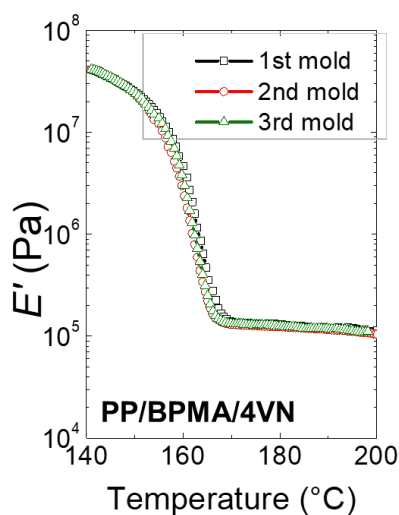


Figure S9. Tensile storage modulus (E') as a function of temperature for 1st, 2nd, and 3rd mold samples of PP/BPMA/4VN, showing full recovery of cross-link density within experimental error after three reprocessing steps.

Table S2. Thermomechanical and thermal properties of 1st, 2nd, and 3rd mold samples of PP CANs

Sample	Mold	$E' @ 180\text{ }^{\circ}\text{C}$ (MPa) ^a	$\tan \delta @$ 180 $^{\circ}\text{C}$ ^a	$T_{m,peak}$ ($^{\circ}\text{C}$) ^b	χ_c (%) ^c
PP/BPMA/2VN	1 st mold	0.12 ± 0.02	0.32 ± 0.02	162 ± 1	37 ± 1
	2 nd mold	0.12 ± 0.01	0.33 ± 0.01	161 ± 1	38 ± 1
	3 rd mold	0.12 ± 0.01	0.33 ± 0.02	161 ± 1	38 ± 1
PP/BPMA/0.5DVB	1 st mold	0.06 ± 0.01	0.52 ± 0.03	160 ± 1	33 ± 1
	2 nd mold	0.06 ± 0.01	0.51 ± 0.02	160 ± 1	32 ± 1
	3 rd mold	0.06 ± 0.01	0.51 ± 0.02	160 ± 1	32 ± 1
PP/BPMA/2VN/0.5DVB	1 st mold	0.20 ± 0.03	0.22 ± 0.02	159 ± 1	36 ± 1
	2 nd mold	0.20 ± 0.02	0.23 ± 0.01	159 ± 1	35 ± 1
	3 rd mold	0.20 ± 0.02	0.22 ± 0.03	159 ± 1	35 ± 1

^aDetermined by DMA

^bDetermined from DSC measurements of the melting peak

^cCalculated as the ratio of the latent heat of fusion of the sample to the latent heat of fusion for 100% crystalline PP (207.1 J/g)^{S3}

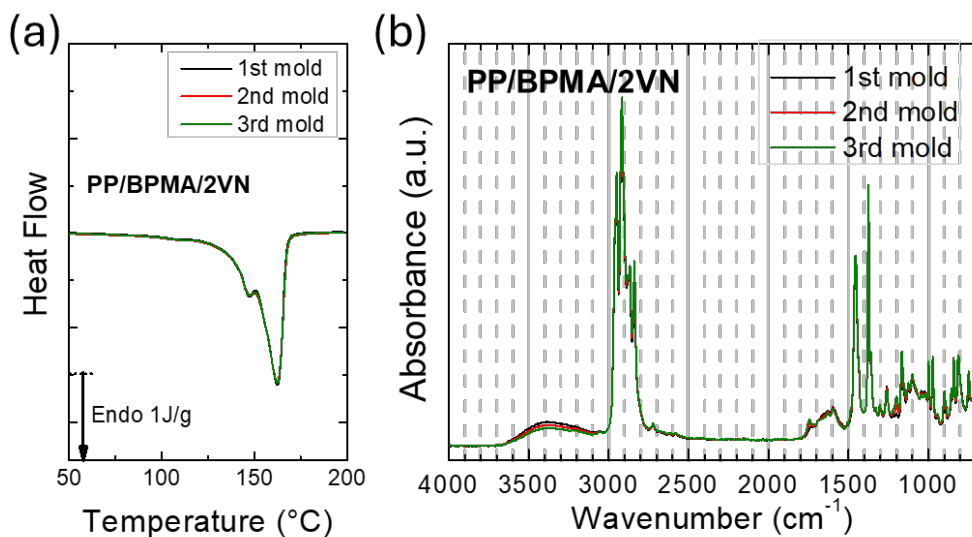


Figure S10. (a) DSC thermograms and (b) FTIR spectra for 1st, 2nd, and 3rd mold samples of PP/BPMA/2VN, indicating complete property recovery.

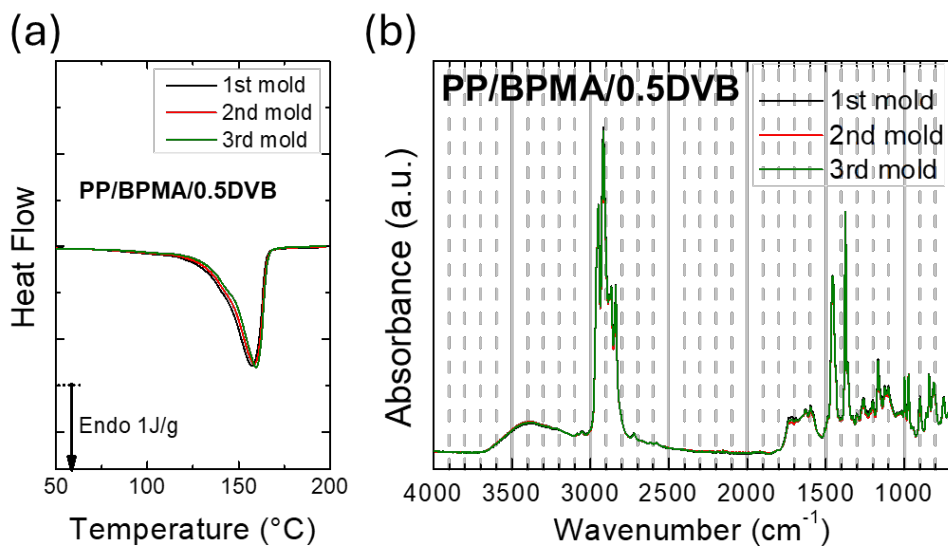


Figure S11. (a) DSC thermograms and (b) FTIR spectra for 1st, 2nd, and 3rd mold samples of PP/BPMA/0.5DVB, indicating complete property recovery.

Table S3. Viscous creep strains of neat, thermoplastic PP sample and 1st mold reactively processed PP samples after 600 s of an applied 3.0 kPa shear stress at 180 °C

Sample	Viscous creep strain (%)
PP	4700 ^a
PP/BPMA	4300 ^a
PP/BPMA/1VN	482 ^b
PP/BPMA/2VN	226 ^b
PP/BPMA/4VN	198 ^b
PP/BPMA/0.5DVB	351 ^b
PP/BPMA/2VN/0.5DVB	117 ^b

^aValues obtained from creep strain at $t = 600$ s

^bValues obtained by extrapolating the slope of the best-fit line of the data between $t = 500$ s and $t = 600$ s for each respective creep curve back to $t = 0$ s and subtracting this y-intercept from the final strain value at $t = 600$ s

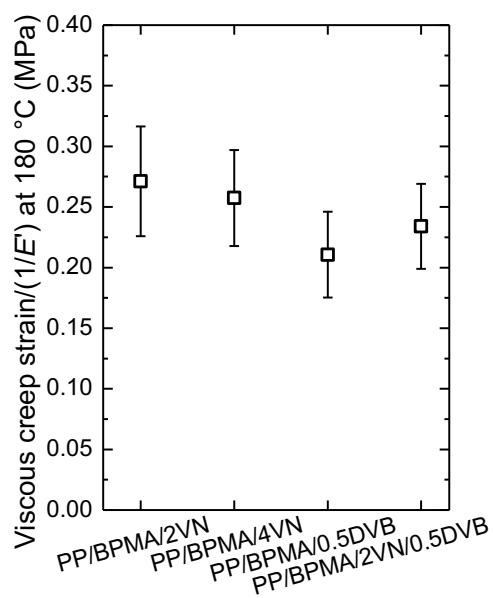


Figure S12. Viscous creep strain normalized by the inverse of E' at 180 °C, the resulting normalized values are invariant within experimental uncertainty across the four sample types, supporting the idea that a higher modulus is correlated with a reduced viscous creep strain.

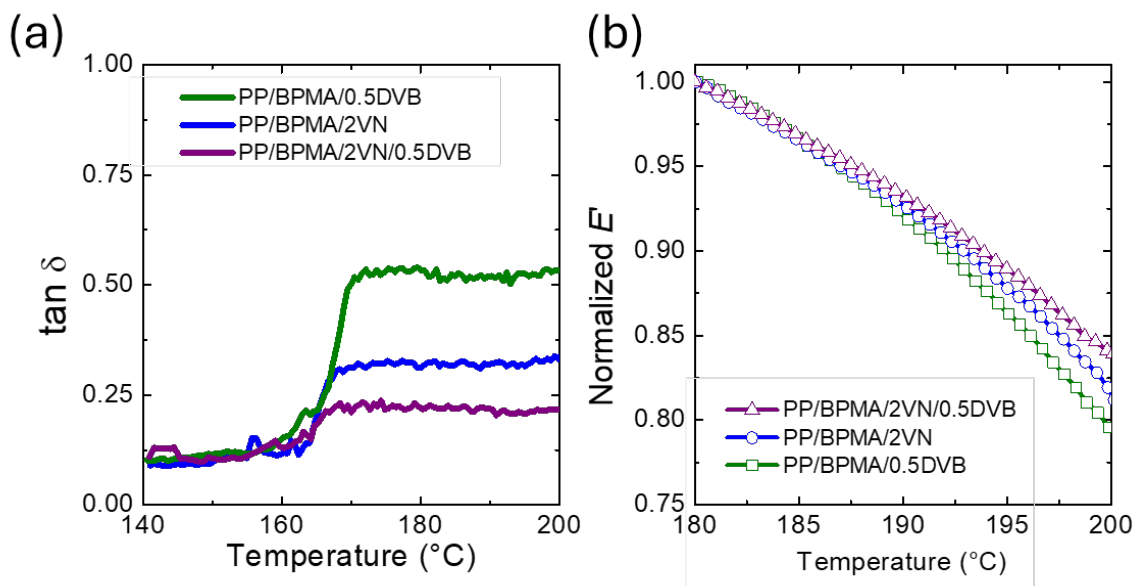


Figure S13. Properties of a 1st mold PP CAN made with both VN and DVB additives (PP/BPMA/2VN/0.5DVB), as compared to samples only made with one or the other: (a) $\tan \delta$ (E''/E') as a function of temperature, (b) normalized E' ($E'(T)$ relative to E' at 180 °C) as a function of temperature in the rubbery plateau.

PP/BPMA/2VN/0.5DVB

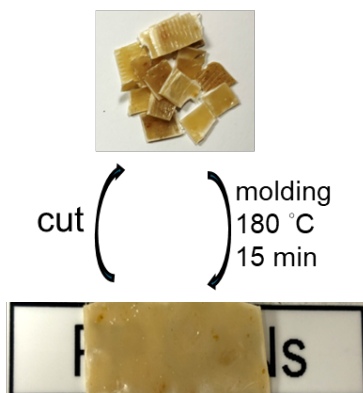


Figure S14. Successful formation of a robust, consolidated 2nd mold sample from cut pieces of a 1st mold sample of PP/BPMA/2VN/0.5DVB.

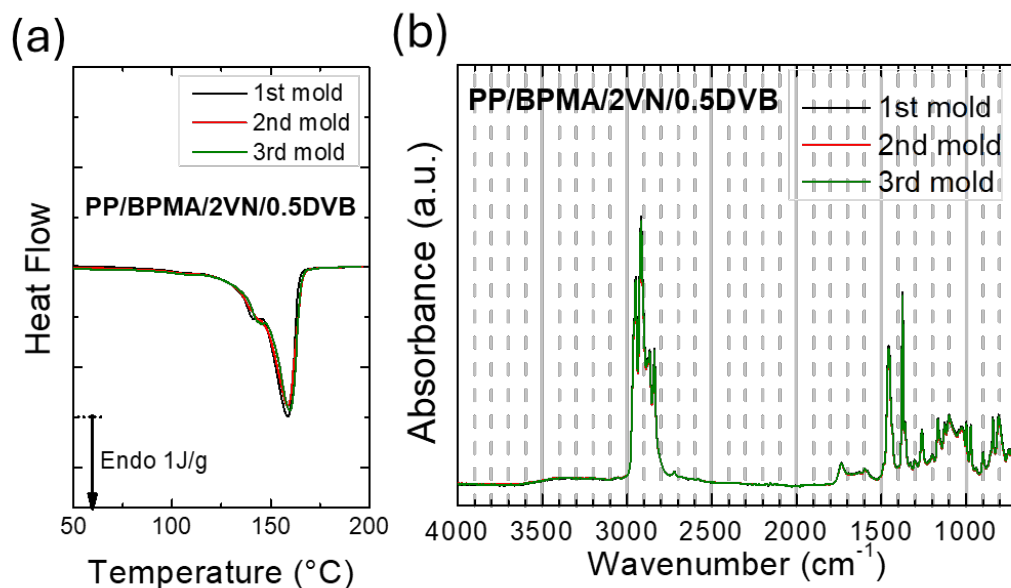


Figure S15. (a) DSC thermograms and (b) FTIR spectra for 1st, 2nd, and 3rd mold samples of PP/BPMA/2VN/0.5DVB, indicating complete property recovery.

REFERENCES

- (S1) Huang, Y.-W.; Suazo, M. J.; Barbon, S. M.; Brown, H. A.; Auyeung, E.; Shan, C. L. P.; Torkelson, J. M. Polypropylene covalent adaptable networks with full cross-link density recovery after reprocessing: Development by free-radical reactive processing with resonance-stabilized, aromatic disulfide cross-linkers. *ACS Macro Lett.* **2025**, *14* (3), 341–348.
- (S2) Dhinojwala, A.; Hooker, J. C.; Torkelson, J. M. Retardation of rotational reorientation dynamics in polymers near the glass transition: a novel study over eleven decades in time using second-order non-linear optics. *J. Non-Cryst. Solids* **1994**, *172*, 286-296.
- (S3) Bu, H. S.; Cheng, S. Z.; Wunderlich, B. Addendum to the thermal properties of polypropylene. *Makromol. Chem., Rapid Commun.* **1988**, *9* (2), 75-77.